Indian Institute of Technology Guwahati

Course Code: EE517
Analog IC Design lab

Experiment 5

Design and analysis of 2-stage OP-AMP

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Contents

1 Objective

The objective of this experiment is to design and analyze a two-stage CMOS operational amplifier (op-amp) using 0.18 μm CMOS process technology, operating from a single 1.8 V supply ($V_{DD} = 1.8$ V, $V_{SS} = 0$ V). The op-amp is required to meet strict performance specifications suitable for high-speed analog integrated circuit applications.

The designed amplifier must achieve a minimum DC open-loop gain of 1000 V/V (60 dB), a unity gain frequency (UGF) of 200 MHz, and a phase margin of at least 60° while driving a 1 pF capacitive load. In addition, the design must ensure low common-mode gain not exceeding 0.1 V/V (≤ -20 dB), symmetrical slew rate performance of 50 V/μs for both rising and falling transitions, and an output differential peak-to-peak swing of at least 1 V_{pp}.

Finally, the amplifier must satisfy a strict low-power constraint with total power consumption limited to 1 mW. The report aims to present the complete design methodology, transistor-level sizing, biasing strategy, frequency compensation approach, and simulation-based verification demonstrating compliance with all given specifications.

2 Theory of Two-Stage CMOS Operational Amplifier Design

A two-stage CMOS operational amplifier is one of the most widely used op-amp topologies in analog integrated circuit design due to its ability to provide high voltage gain, moderate output swing, and sufficient drive capability for capacitive loads. The two-stage architecture is especially suitable for low-voltage CMOS processes such as 0.18 μm technology, where a single gain stage may not provide sufficient open-loop gain.

2.1 Two-Stage Op-Amp Architecture

A typical two-stage CMOS op-amp consists of:

- **First Stage (Input Differential Stage):** Provides initial voltage gain and ensures high input impedance.
- **Second Stage (Gain/Output Stage):** Provides additional voltage gain and improves output drive capability.

The block-level structure is shown as:

$$A_v = A_{v1} \times A_{v2} \tag{1}$$

where A_{v1} is the gain of the first stage and A_{v2} is the gain of the second stage.

2.2 First Stage: Differential Amplifier

The first stage is usually implemented using a MOS differential pair with an active current mirror load. The differential pair converts the differential input voltage into a single-ended output current, which is then translated into a voltage at the output node.

The small-signal voltage gain of the first stage is given by:

$$A_{v1} \approx g_{m1} \left(r_{o1} \parallel r_{o3} \right) \tag{2}$$

where:

- g_{m1} is the transconductance of the input MOS transistor,
- r_{o1} and r_{o3} are the output resistances of the differential pair transistor and load transistor respectively.

The differential input stage provides high common-mode rejection ratio (CMRR) and sets the input common-mode operating range of the op-amp.

2.3 Second Stage: Common Source Gain Stage

The second stage is generally a common-source amplifier with an active load. This stage provides high voltage gain and drives the load capacitance.

The gain of the second stage is approximately:

$$A_{v2} \approx g_{m6} \left(r_{o6} \parallel r_{o7}\right) \tag{3}$$

where:

- g_{m6} is the transconductance of the second stage transistor,
- r_{o6} and r_{o7} represent the output resistances of the amplifying transistor and load transistor.

2.4 Overall DC Gain

The overall DC open-loop gain of the two-stage op-amp is given by:

$$A_{v0} = A_{v1} \times A_{v2} \tag{4}$$

In decibels (dB), the gain is expressed as:

$$A_{v0(dB)} = 20 \log_{10}(A_{v0}) \tag{5}$$

To satisfy high gain requirements (≥ 60 dB), both stages must provide sufficient gain, typically in the range of 30–40 dB each.

2.5 Frequency Response and Stability

Since the two-stage amplifier contains two high-gain nodes, it naturally introduces at least two dominant poles. Without compensation, the phase shift may reach 180° before the gain crosses unity, leading to instability and oscillations.

The open-loop transfer function of a two-stage op-amp can be approximated as:

$$A(s) = \frac{A_0 \left(1 + \frac{s}{\omega_z}\right)}{\left(1 + \frac{s}{\omega_{p1}}\right)\left(1 + \frac{s}{\omega_{p2}}\right)} \tag{6}$$

where p_1 and p_2 are the dominant and non-dominant poles.

2.6 Miller Compensation

To ensure closed-loop stability, Miller compensation is commonly employed using a compensation capacitor C_c connected between the output of the second stage and the output of the first stage.

Miller compensation increases the effective capacitance at the first-stage output node by the Miller multiplication factor:

$$C_M = C_c(1 + A_{v2}) \tag{7}$$

This shifts the dominant pole to a lower frequency:

$$p_1 \approx \frac{1}{R_1 C_M} \tag{8}$$

where R_1 is the effective resistance at the first stage output.

The second pole is typically located at:

$$p_2 \approx \frac{1}{R_2 C_L} \tag{9}$$

where R_2 is the output resistance of the second stage and C_L is the load capacitance.

2.7 Unity Gain Frequency (UGF)

The unity gain frequency is the frequency at which the magnitude of open-loop gain becomes 1 (0 dB). For a Miller compensated two-stage op-amp, the unity gain frequency is approximated as:

$$\omega_u \approx \frac{g_{m1}}{C_c} \tag{10}$$

In terms of frequency:

$$f_u = \frac{\omega_u}{2\pi} \approx \frac{g_{m1}}{2\pi C_c} \tag{11}$$

Thus, to achieve a higher UGF, either a larger g_{m1} or a smaller C_c is required.

2.8 Phase Margin

Phase margin (PM) is a key stability parameter defined as:

$$PM = 180^\circ + \angle A(j\omega_u) \tag{12}$$

For stable operation with minimal ringing, the phase margin is generally designed to be greater than 60°. Proper selection of C_c and the placement of the non-dominant pole are critical for meeting this requirement.

2.9 Slew Rate

The slew rate is the maximum rate of change of the output voltage when the op-amp is subjected to a large step input. In a two-stage op-amp with Miller compensation, the slew rate is limited by the current charging or discharging the compensation capacitor C_c.

The positive (rising) slew rate is given by:

$$SR_+ = \frac{I_{bias}}{C_c} \tag{13}$$

The negative (falling) slew rate is given by:

$$SR_- = \frac{I_{sink}}{C_c} \tag{14}$$

To achieve symmetrical slew rate performance, the sourcing and sinking currents must be properly matched.

2.10 Output Voltage Swing

The output voltage swing is limited by transistor saturation requirements in the output stage. For proper amplification, the output transistors must remain in saturation. The maximum and minimum output voltages can be approximated as:

$$V_{out,max} \approx V_{DD} - V_{DSsat,p} \tag{15}$$

$$V_{out,min} \approx V_{SS} + V_{DSsat,n} \tag{16}$$

Thus, the differential peak-to-peak output swing is limited by headroom constraints in low-voltage CMOS processes.

2.11 Power Consumption

The total power dissipation of the op-amp is primarily due to the DC bias currents. It is expressed as:

$$P_{diss} = (V_{DD} - V_{SS}) \times I_{total} \tag{17}$$

To satisfy low power requirements, the bias currents must be carefully selected while still maintaining the required gain, bandwidth, and slew rate specifications.

2.12 Advantages of Adding a Series Resistance with the Compensation Capacitor

In a two-stage CMOS operational amplifier, Miller compensation is commonly implemented using a compensation capacitor C_c connected between the output of the first stage and the output of the second stage. However, this configuration introduces an undesired right-half plane (RHP) zero, which reduces the phase margin and degrades the stability of the amplifier. To overcome this limitation, a resistor R_z is often connected in series with the compensation capacitor.

The advantages of adding a series resistance R_z with the compensation capacitor C_c are given below:

- **Elimination of the Right-Half Plane (RHP) Zero:** In conventional Miller compensation, the feedforward path through C_c introduces a RHP zero. By adding R_z, this zero can be shifted from the right-half plane to the left-half plane (LHP), improving stability.

- **Improved Phase Margin:** Since the RHP zero contributes additional negative phase shift, it reduces the phase margin. Introducing R_z moves the zero to a lower-frequency LHP location, contributing positive phase lead and thus improving the phase margin.

- **Enhanced Frequency Compensation and Stability:** The resistor R_z allows better pole-zero placement, making the amplifier easier to stabilize for large capacitive loads and high unity gain frequency operation.

- **Reduction of Gain Peaking:** Without R_z, the phase degradation may cause gain peaking near the unity gain frequency, leading to overshoot and ringing in the transient response. The series resistor helps reduce this peaking and improves transient behavior.

- **Better Settling Response:** By improving stability and reducing ringing, the op-amp achieves faster settling time and improved dynamic performance, which is essential in high-speed applications.

- **Flexibility in Compensation Design:** The value of R_z provides an additional degree of freedom for tuning the frequency response. This enables designers to optimize the trade-off between bandwidth, phase margin, and stability.

Thus, adding a series resistance R_z along with the compensation capacitor C_c is an effective technique to improve the stability, phase margin, and transient performance of a two-stage CMOS operational amplifier.

2.13 Summary

In summary, the two-stage CMOS operational amplifier provides high gain by cascading a differential input stage with a common-source gain stage. However, stability becomes a major design challenge due to the presence of multiple poles. Miller compensation is widely used to ensure adequate phase margin. Key design trade-offs exist between gain, bandwidth, slew rate, output swing, and power consumption, and transistor sizing must be optimized to meet all performance requirements.

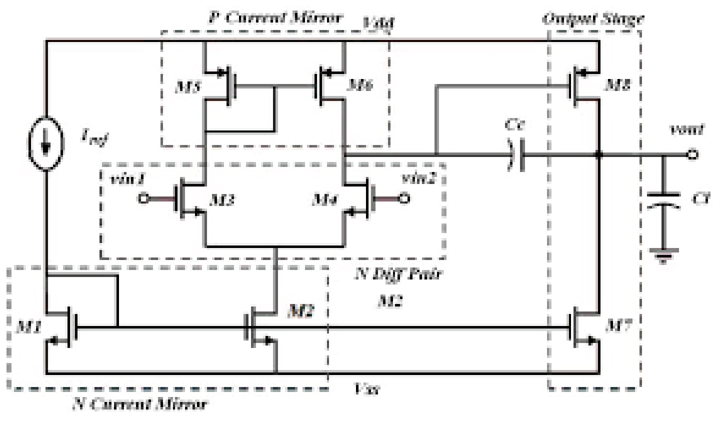

Figure 1: Two stage Op-AMP with only coupling capacitor

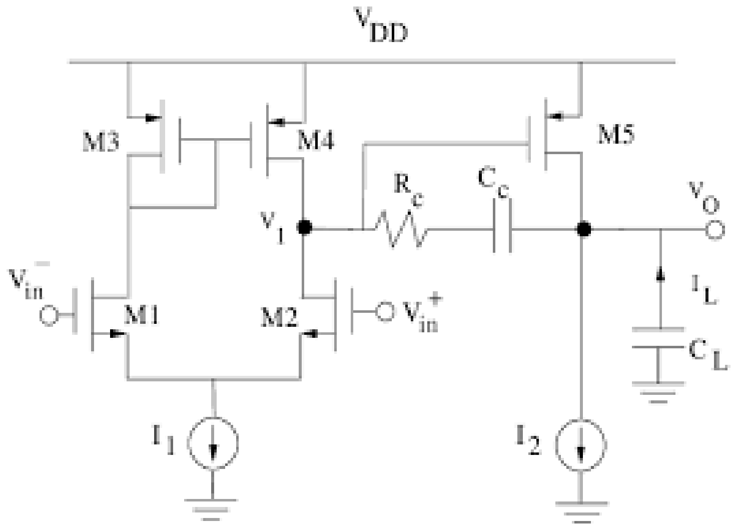

Figure 2: Two stage Op-AMP with coupling capitor and resistor

3 Sizing of transistor and capacitance to create a op amp of required specification

We size our coupling capacitance to get required value of phase margin.

We decide the value of our reference current source from our required slew rate.

We decide sizing for Tx M3 and M4 from our unity gain frequency or Gain bandwidth product.

We decide sizing for Tx M5 and M6 from our Input common mode range (high) or ICMR+, we assume it to be 1.6 V, for this we will also need to find minimum threshold voltage for Tx M3 and max threshold voltage for Tx M5. We find it using simulation.

We decide sizing of Tx M2 from out ICMR- (we assume it to be 0.8V), for this we will need max threshold voltage of Tx M3. again to find this we perform simulation. Using this we will find the minimum Vds required to keep our Tx M2 in saturation, using this minimum overdrive voltage we will find our required size for Tx M2 using current voltage equation for MOSFET(we found this current earlier)

We decide size for our Tx M8 from Phase Margin , we require $gm_{M6} > 10 * gm_{M3}$ using this relation and current mirror property we will find size for Tx M8. Here we also find current flowing in Tx M8.

Now since we know current that need to flow through M8 and M7 we can size our Tx M7 to make the required current flow through M8 as M7 forms a current mirror.

Similarly we do for our second 2 stage op amp circuit with resitance in series with coupling capacitor.

4 Results

4.1 2-Stage OP AMP with only coupling capacitance

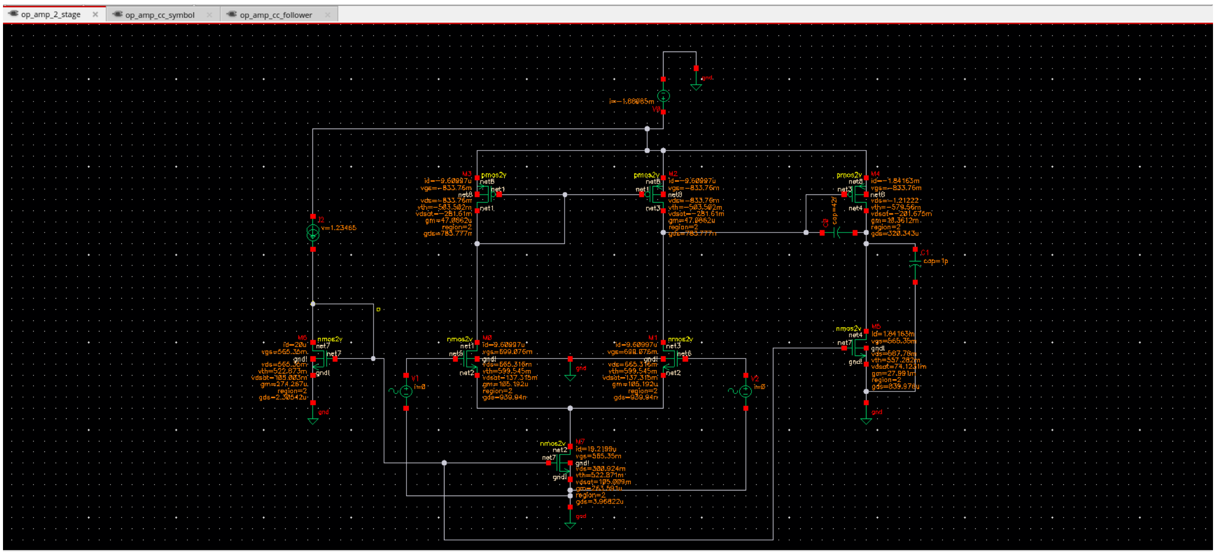

Figure 3: 2 stage Op amp with only Cc

This is our circuit for 2 stage op amp using only coupling capacitance Cc as we can see all the transistors are in saturation (region 2) which means all the transistors are biased properly.

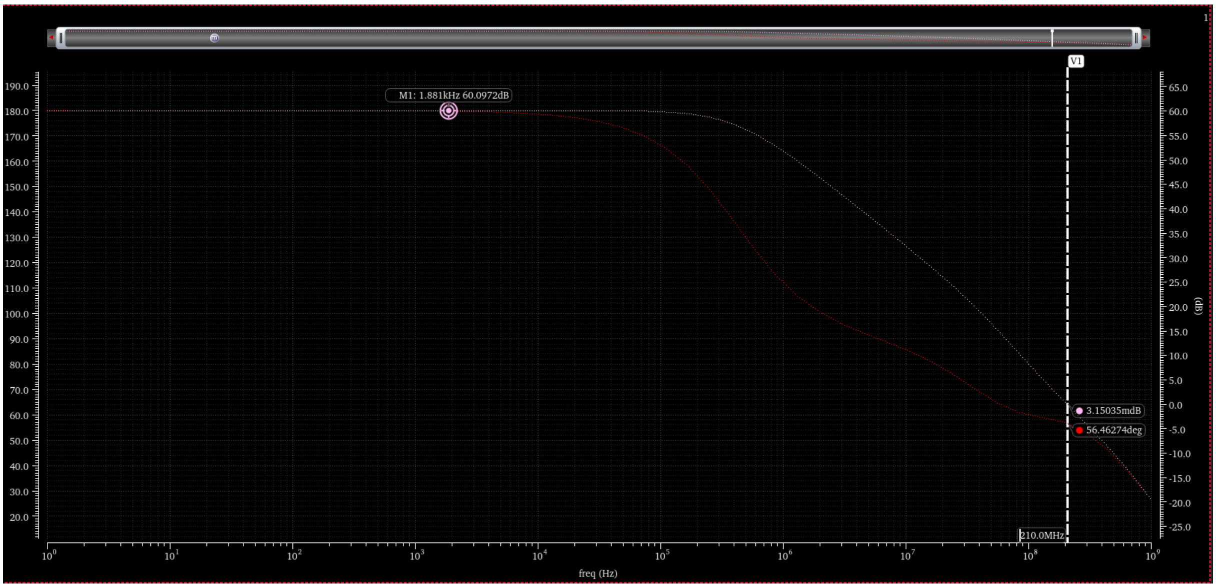

Figure 4: Gain frequency and Phase plot

The above figure shows the gain and phase response of the designed two-stage operational amplifier with Miller (coupling) compensation. The low-frequency open-loop gain is approximately 60 dB, satisfying the required DC gain specification. This high gain is achieved through cascading two high-gain amplifier stages.

The unity gain frequency (UGF) is observed at approximately 210 MHz, which meets the target bandwidth requirement of 200 MHz. The UGF is primarily determined by the transconductance of the input stage and the value of the compensation capacitor, as given by

$$f_u \approx \frac{g_{m1}}{2\pi C_c}. \tag{18}$$

The phase margin is measured to be approximately 56.5°, which is close to the desired value of 60°. This indicates a stable closed-loop operation with minimal overshoot and good transient response. The dominant pole introduced by the Miller capacitor ensures pole splitting, pushing the non-dominant poles to higher frequencies and improving stability.

Overall, the frequency response confirms that the amplifier meets the required specifications of high gain, wide bandwidth, and adequate phase margin, validating the effectiveness of Miller compensation in achieving stable high-speed operation.

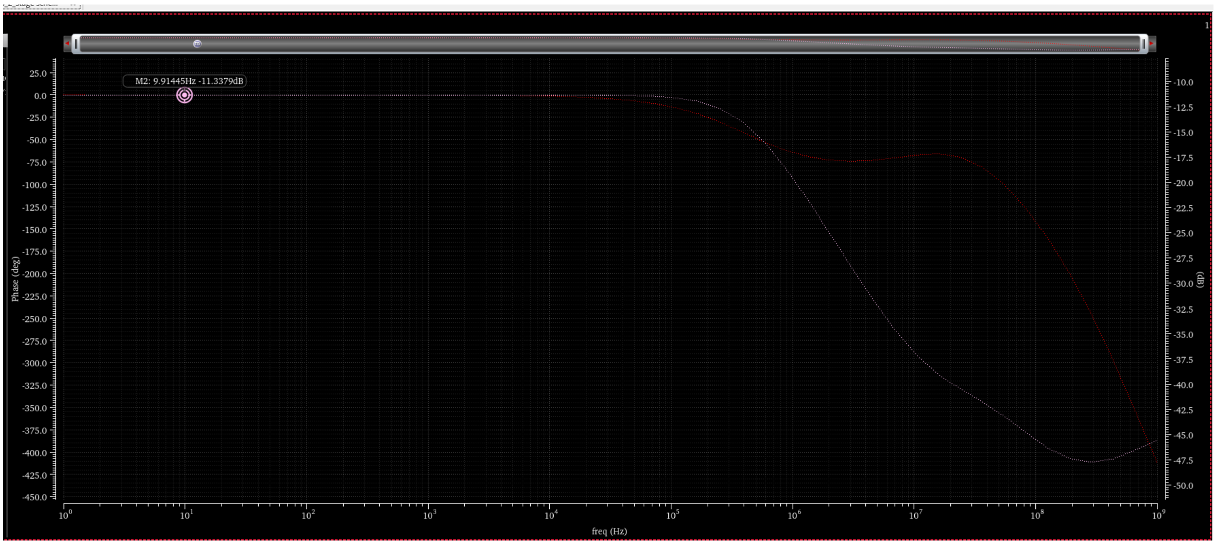

Figure 5: Common Mode gain

The above figure illustrates the frequency response of the common-mode gain (A_{cm}) of the two-stage operational amplifier with Miller (coupling) compensation. The low-frequency common-mode gain is approximately -11.3 dB, indicating effective suppression of common-mode signals, primarily due to the differential input stage and high output resistance of the tail current source.

At low frequencies, the high common-mode rejection is achieved through symmetric circuit operation and current source biasing. As frequency increases, the common-mode gain magnitude decreases further due to the dominant pole introduced by the compensation capacitor and intrinsic device capacitances. This behavior improves high-frequency common-mode noise attenuation.

The observed roll-off in A_{cm} enhances the overall common-mode rejection ratio (CMRR) across the operating bandwidth, contributing to improved noise immunity and signal integrity. The frequency response confirms the effectiveness of the compensation scheme in maintaining low common-mode gain while preserving amplifier stability.

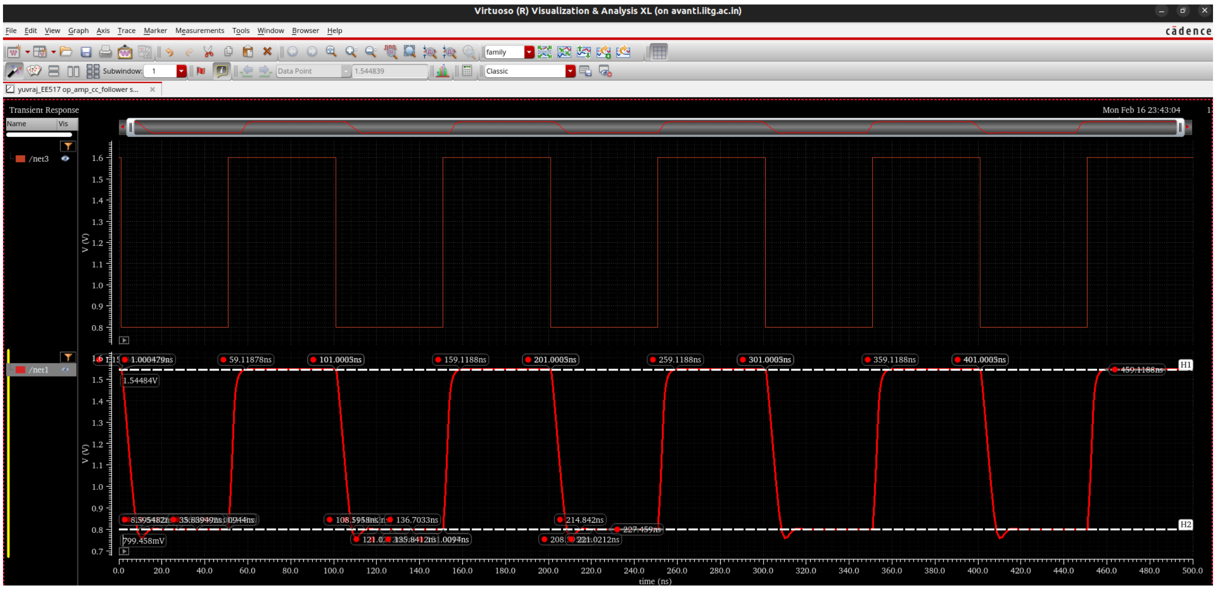

Figure 6: Pulse input with 10Mhz

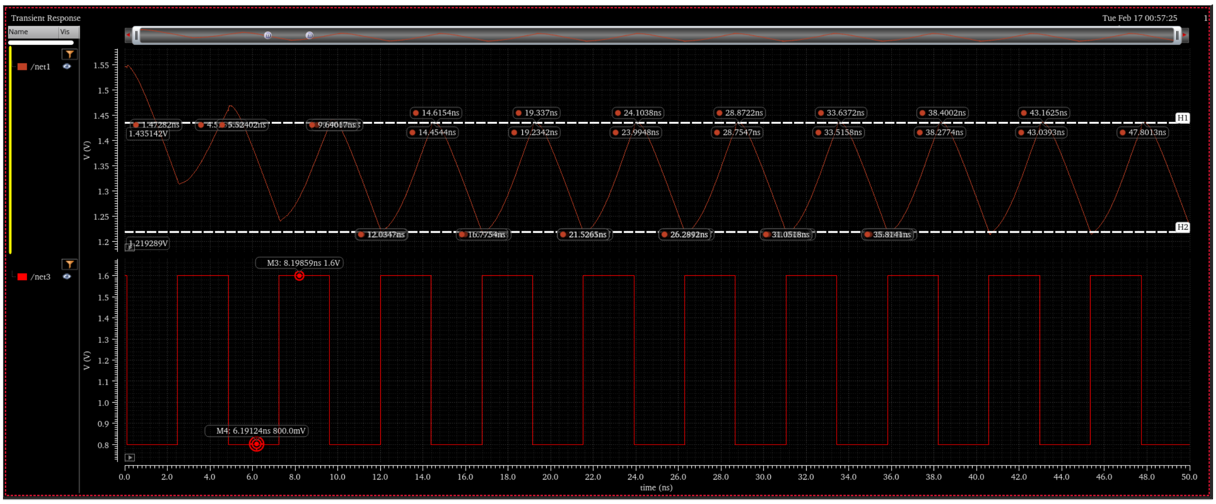

Figure 7: Pulse input with 210 Mhz

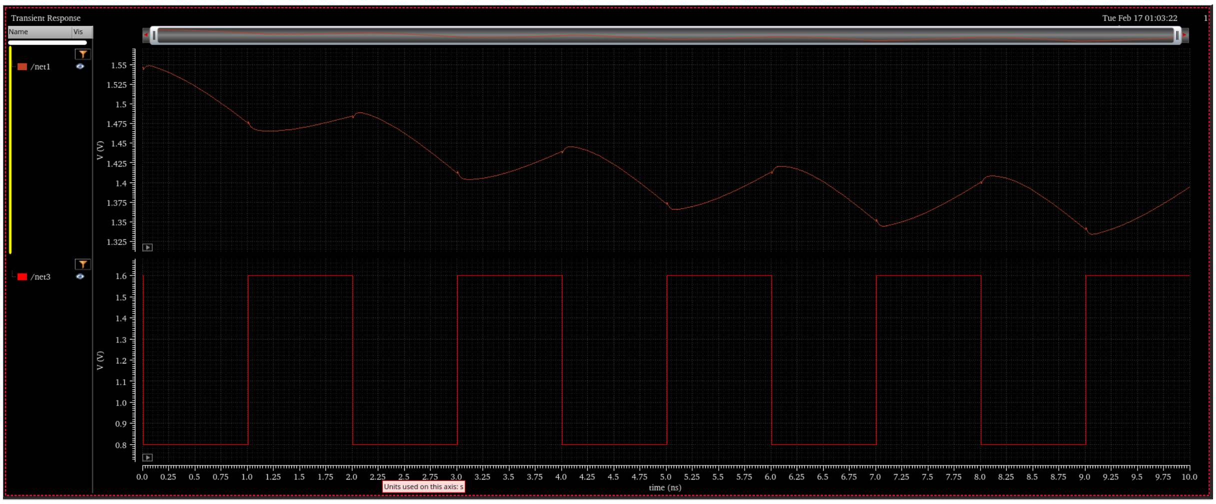

Figure 8: Pulse input with 500 Mhz

The transient performance of the two-stage operational amplifier configured as a voltage follower with Miller compensation was evaluated using a pulse input of varying frequencies: 10 MHz, 210 MHz, and 500 MHz. The slew rate (SR), defined as the maximum rate of change of the output voltage, is given by

$$\mathrm{SR} = \max \left| \frac{dV_{out}}{dt} \right| . \tag{19}$$

At 10 MHz, the amplifier exhibits a high and nearly constant slew rate of approximately $+180$ V/μs for the rising edge and -110 V/μs for the falling edge, indicating that the output transitions are primarily limited by the bias currents charging and discharging the compensation capacitor. At 210 MHz, the slew rate reduces to approximately $+109$ V/μs and -106 V/μs, reflecting bandwidth and current limitations as the signal frequency approaches the unity gain frequency.

At 500 MHz, a constant slew rate is no longer observed, and the output waveform becomes significantly distorted. This behavior is attributed to severe slew-rate limiting and insufficient settling time, as the required output voltage transition exceeds the maximum charging and discharging capability of the internal nodes. Additionally, parasitic capacitances and higher-order poles dominate the transient response, leading to nonlinear and frequency-dependent output slopes.

This analysis highlights the large-signal dynamic limitations of the amplifier and demonstrates that reliable linear operation is maintained up to frequencies near the unity gain bandwidth, beyond which severe distortion occurs. Hence, transient and slew-rate characterization is critical for determining the maximum usable operating frequency and ensuring signal integrity in high-speed analog application.

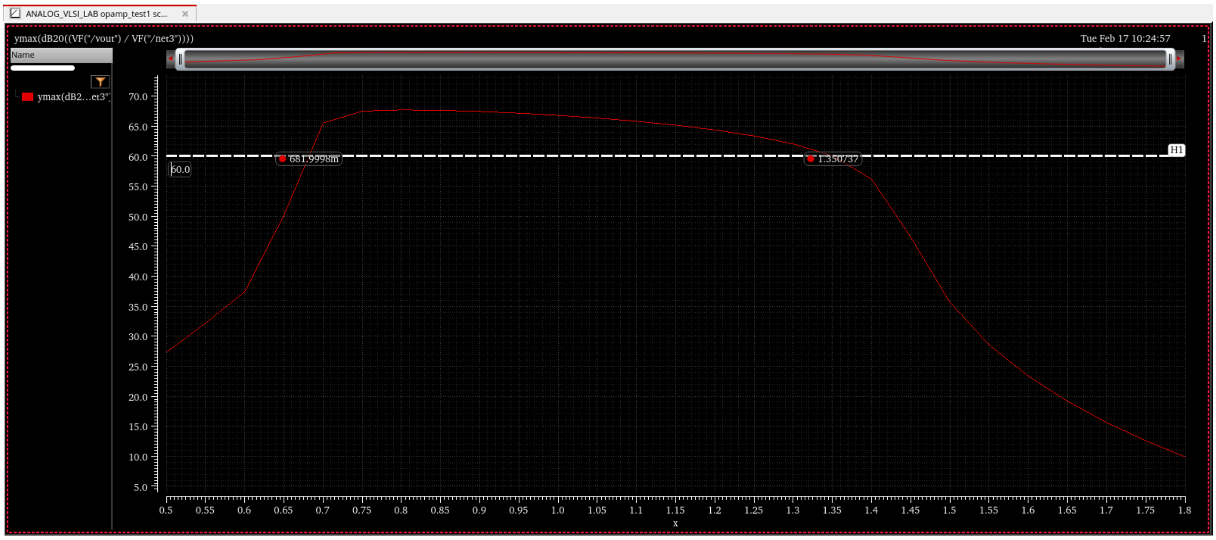

Figure 9: ICMR for 2 stage op amp with coupling capacitance

The input common-mode range (ICMR) defines the allowable range of DC input voltage over which the operational amplifier maintains proper biasing and stable operation while preserving the required gain. shows the variation of differential gain with respect to the applied common-mode input voltage.

From the plot, the amplifier maintains a gain of approximately 60 dB within the input voltage range of 0.68 V to 1.35 V, which corresponds to the valid ICMR. Below the lower limit, the input differential pair enters the triode region or cuts off, whereas above the upper limit, the current mirror and load devices move out of saturation, resulting in gain degradation.

This analysis verifies that the designed amplifier satisfies the required input swing constraints while ensuring stable biasing and high gain operation. Proper ICMR is essential for maximizing signal swing, preventing distortion, and maintaining linearity in precision analog applications.

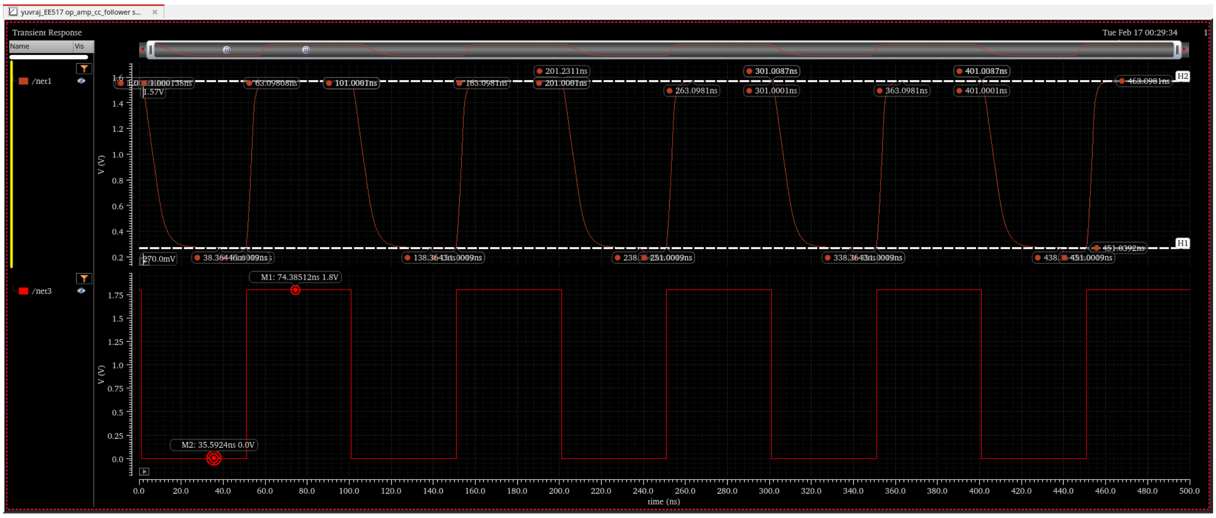

Figure 10: OCMR for 2 stage op amp with coupling capacitance

The output common-mode range (OCMR) defines the maximum allowable voltage swing at the output node for which the operational amplifier maintains linear operation without distortion or saturation. shows the transient response of the amplifier configured as a voltage follower when excited with a pulse input varying from 0 to 1.8 V.

From the observed output waveform, the amplifier is able to swing between approximately 0.27 V and 1.57 V, which represents the valid OCMR. The lower and upper swing limitations arise due to saturation of the output-stage transistors and headroom constraints imposed by the bias currents and device overdrive voltages.

This analysis confirms that the amplifier supports a wide output voltage swing within the supply limits, ensuring low distortion and reliable large-signal operation. Proper OCMR is critical for maximizing dynamic range and maintaining linearity in high-performance analog circuits.

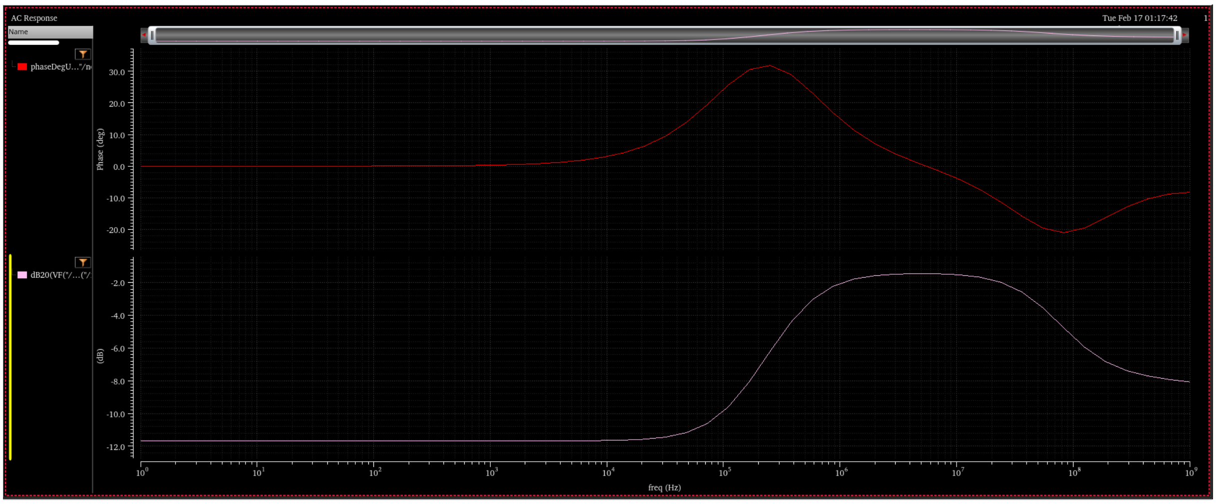

Figure 11: PSRR for 2 stage op amp with coupling capacitance

The power supply rejection ratio (PSRR) characterizes the ability of an operational amplifier to suppress variations in the supply voltage from appearing at the output. It is defined as

$$\mathrm{PSRR}(f) = 20\log_{10}\left(\frac{A_d(f)}{A_{ps}(f)}\right), \tag{20}$$

where $A_d(f)$ is the differential gain and $A_{ps}(f)$ is the gain from supply to output. In this analysis, a 10 mV AC signal is applied at V_{DD}, and the resulting output response is used to compute the common-mode gain, from which PSRR is derived.

At low frequencies, the PSRR is high due to the strong isolation provided by the biasing current sources and high output resistance of the active devices, effectively attenuating supply noise. As frequency increases, parasitic capacitances and finite output impedance introduce direct coupling paths between the supply and output nodes, resulting in degradation of PSRR. The mid-frequency peak observed in the response is attributed to pole–zero interactions introduced by the compensation capacitor and internal node capacitances.

At high frequencies, the PSRR further deteriorates due to reduced loop gain and dominant capacitive feedthrough, allowing more supply noise to propagate to the output. This behavior highlights the inherent trade-off between bandwidth and supply noise rejection in high-speed operational amplifier design and emphasizes the importance of compensation and layout optimization for robust power supply immunity.

4.2 2-Stage OP AMP with coupling capacitance and resistance

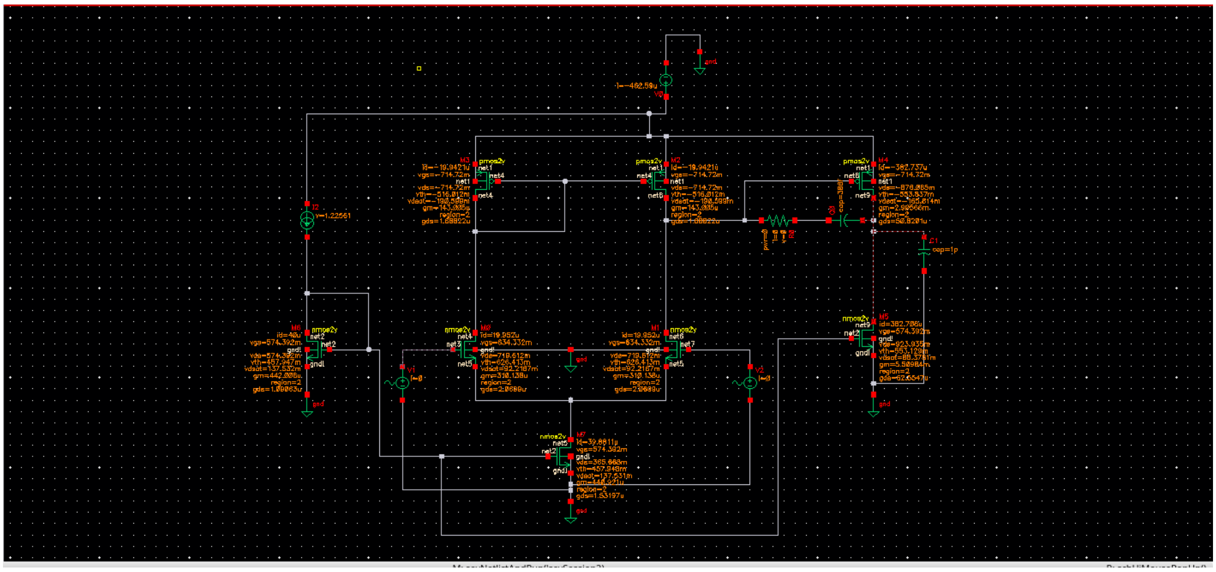

Figure 12: 2 stage Op amp with R and Cc

This is our circuit for 2 stage op amp using resistance and coupling capacitance Cc, as we can see all the transistors are in saturation (region 2) which means all the transistors are biased properly.

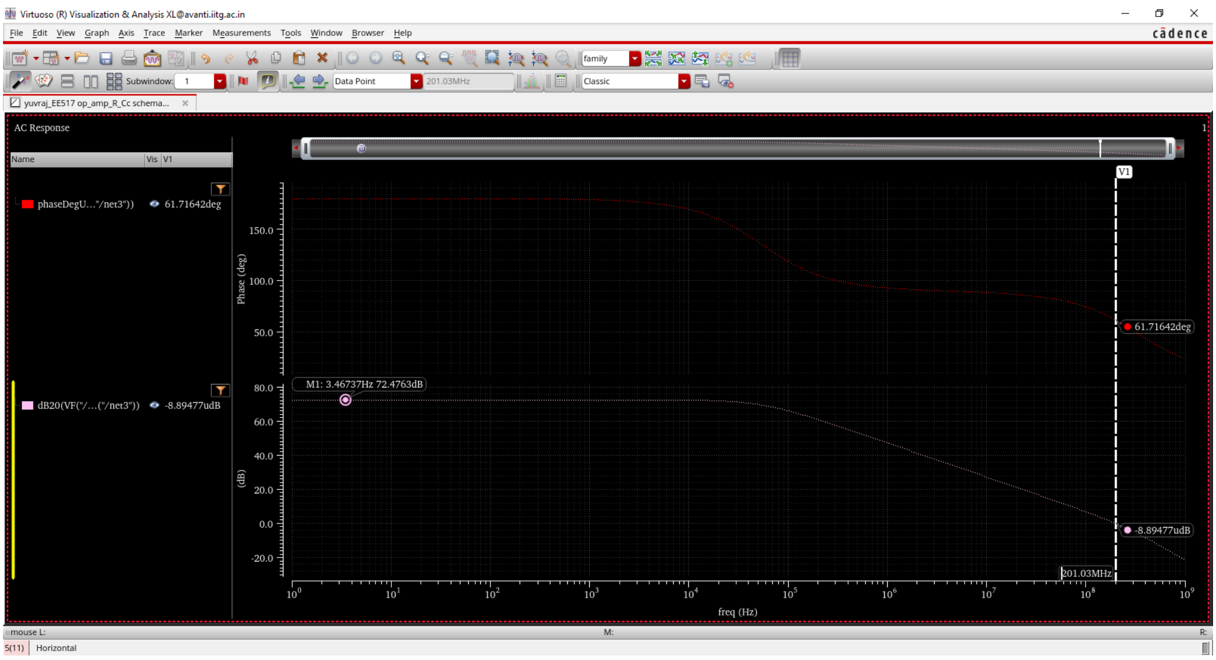

Figure 13: Gain freq and Phase plot

The above figure shows the gain and phase response of the designed two-stage operational amplifier with Miller (coupling) compensation. The low-frequency open-loop gain is 72.4236 dB, satisfying the required DC gain specification. This high gain is achieved through cascading two high-gain amplifier stages.

The unity gain frequency (UGF) is observed at approximately 201.03 MHz, which meets the target bandwidth requirement of 200 MHz. The UGF is primarily determined by the transconductance of the input stage and the value of the compensation capacitor, as given by

$$f_u \approx \frac{g_{m1}}{2\pi C_c}. \tag{21}$$

The phase margin is measured to be approximately 61.7164°, which is greater than desired value of 60°. This indicates a stable closed-loop operation with minimal overshoot and good transient response. The dominant pole introduced by the Miller capacitor and resistor ensures pole splitting, pushing the non-dominant poles to higher frequencies and improving stability.

Overall, the frequency response confirms that the amplifier meets the required specifications of high gain, wide bandwidth, and adequate phase margin, validating the effectiveness of Miller compensation in achieving stable high-speed operation.

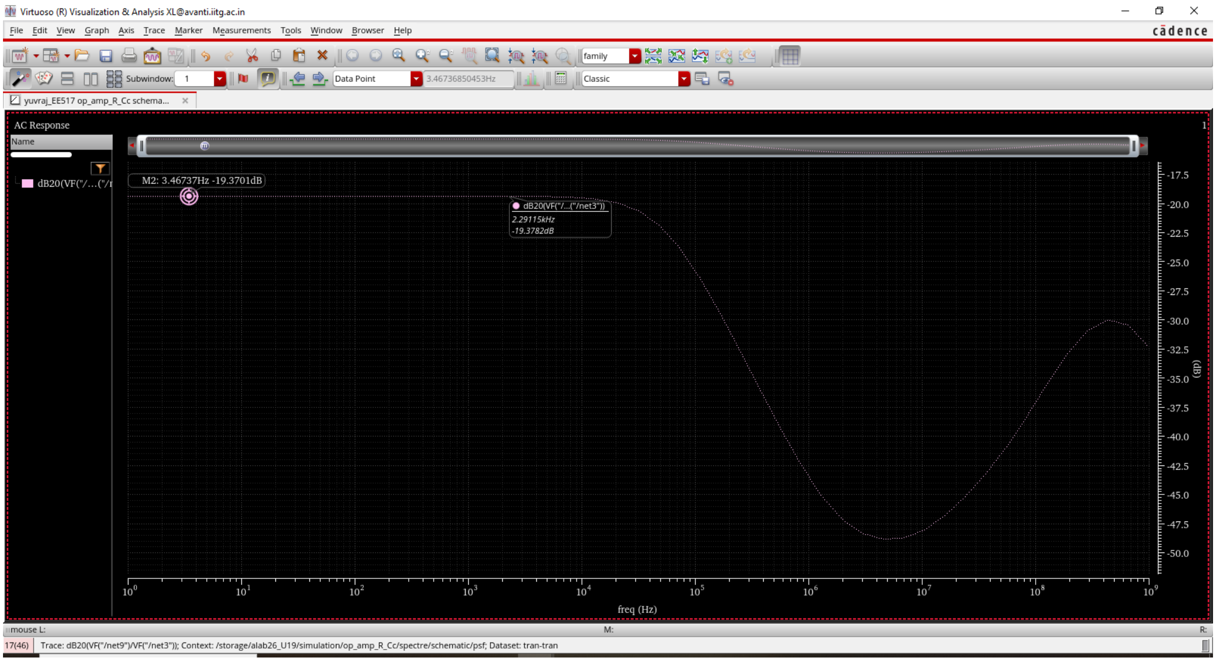

Figure 14: Common mode gain

The above figure illustrates the frequency response of the common-mode gain (A_{cm}) of the two-stage operational amplifier with Miller (coupling) compensation. The low-frequency common-mode gain is approximately −19.370 dB, indicating effective suppression of common-mode signals, primarily due to the differential input stage and high output resistance of the tail current source.

At low frequencies, the high common-mode rejection is achieved through symmetric circuit operation and current source biasing. As frequency increases, the common-mode gain magnitude decreases further due to the dominant pole introduced by the compensation capacitor and intrinsic device capacitances. This behavior improves high-frequency common-mode noise attenuation.

The observed roll-off in A_{cm} enhances the overall common-mode rejection ratio (CMRR) across the operating bandwidth, contributing to improved noise immunity and signal integrity. The frequency response confirms the effectiveness of the compensation scheme in maintaining low common-mode gain while preserving amplifier stability.

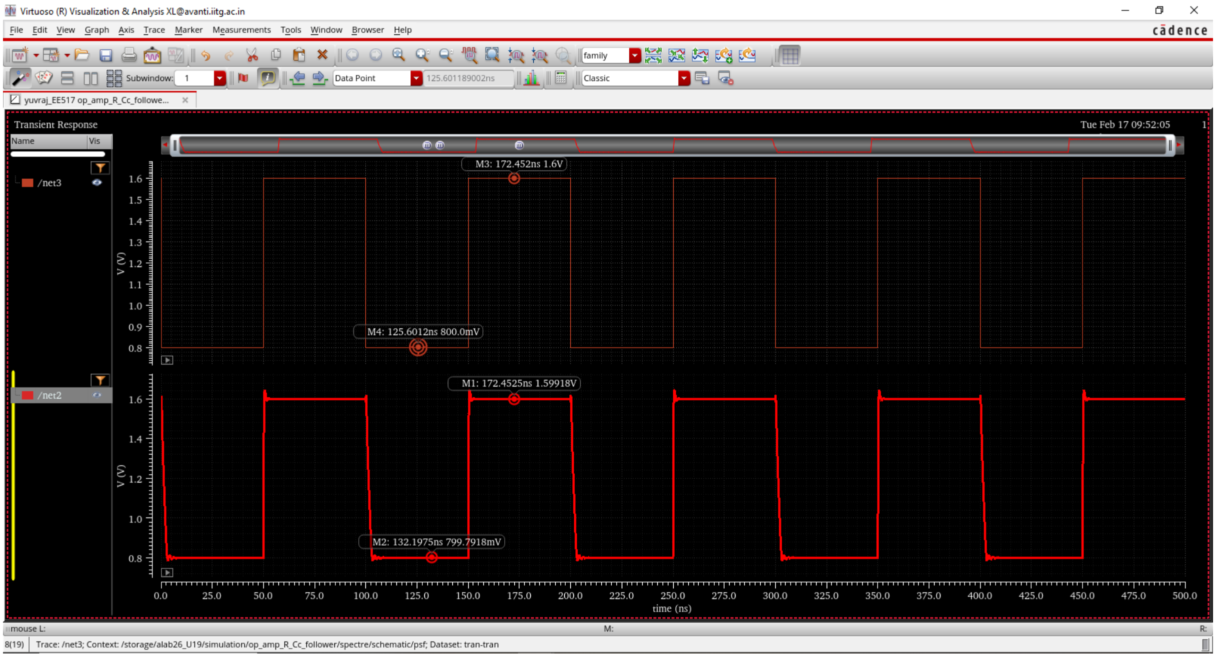

Figure 15: Pulse input with 10 Mhz

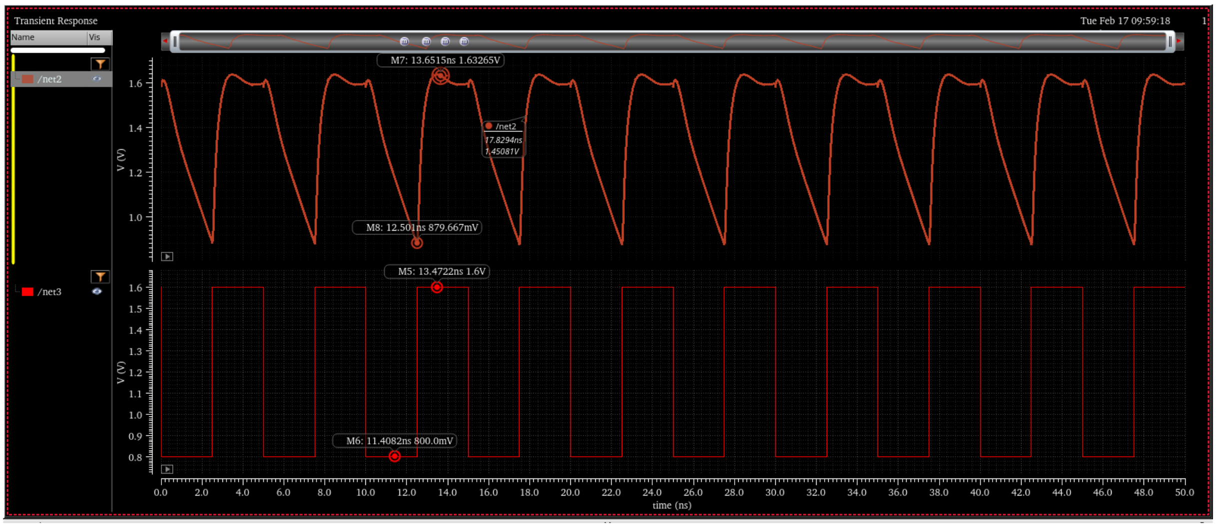

Figure 16: Pulse input with 200 Mhz

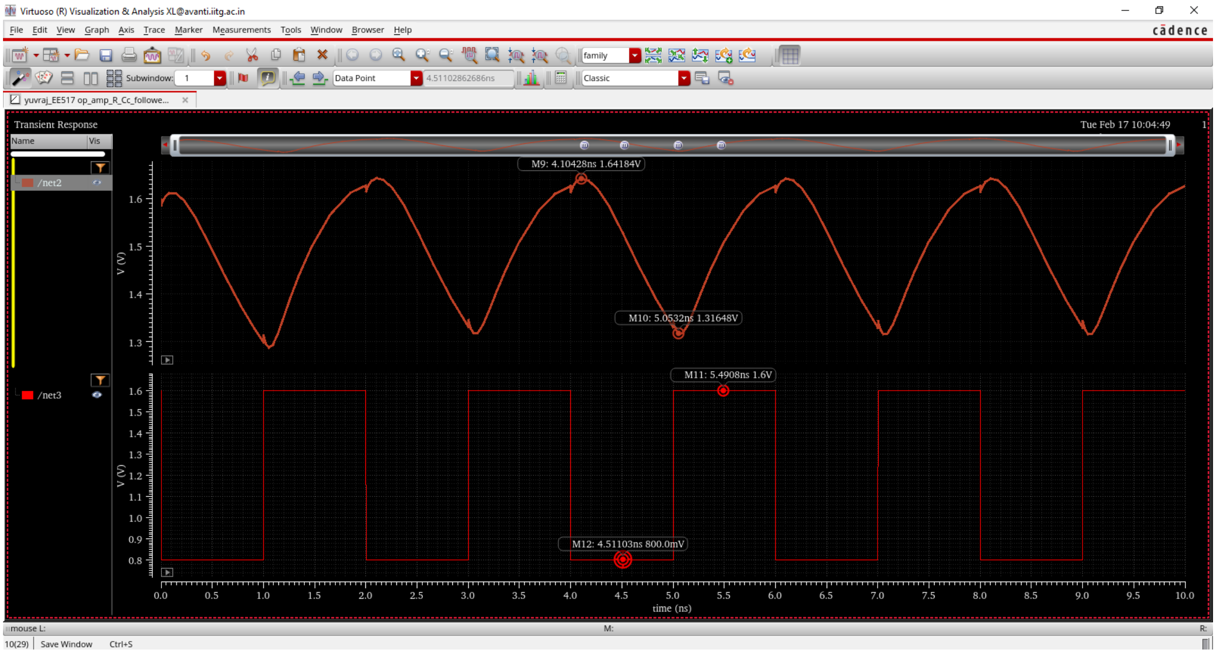

Figure 17: Pulse input with 500 Mhz

The transient performance of the two-stage operational amplifier configured as a voltage follower with Miller compensation was evaluated using a pulse input of varying frequencies: 10 MHz, 200 MHz, and 500 MHz. The slew rate (SR), defined as the maximum rate of change of the output voltage, is given by

$$\mathrm{SR} = \max \left| \frac{dV_{out}}{dt} \right| . \tag{22}$$

At 10 MHz, the amplifier exhibits a high and nearly constant slew rate of approximately $+2560$ V/μs for the rising edge and -300 V/μs for the falling edge, indicating that the output transitions are primarily limited by the bias currents charging and discharging the compensation capacitor. At 200 MHz, the slew rate reduces to approximately $+1391$ V/μs and -323 V/μs, reflecting bandwidth and current limitations as the signal frequency approaches the unity gain frequency.

At 500 MHz, a constant slew rate is observed which did not happen in our previous Op amp circuit, this show that including a resistance with our coupling capacitance massively boosted our phase response of our circuit and its stability at higher frequency. Our slew rate for rising edge and falling edge comes out to be $+388$ V/μs and -401.6 V/μs respectively.

This analysis highlights the large-signal dynamic limitations of the amplifier and demonstrates that reliable linear operation is maintained up to frequencies near the unity gain bandwidth, beyond which severe distortion occurs. Hence, transient and slew-rate characterization is critical for determining the maximum usable operating frequency and ensuring signal integrity in high-speed analog application.

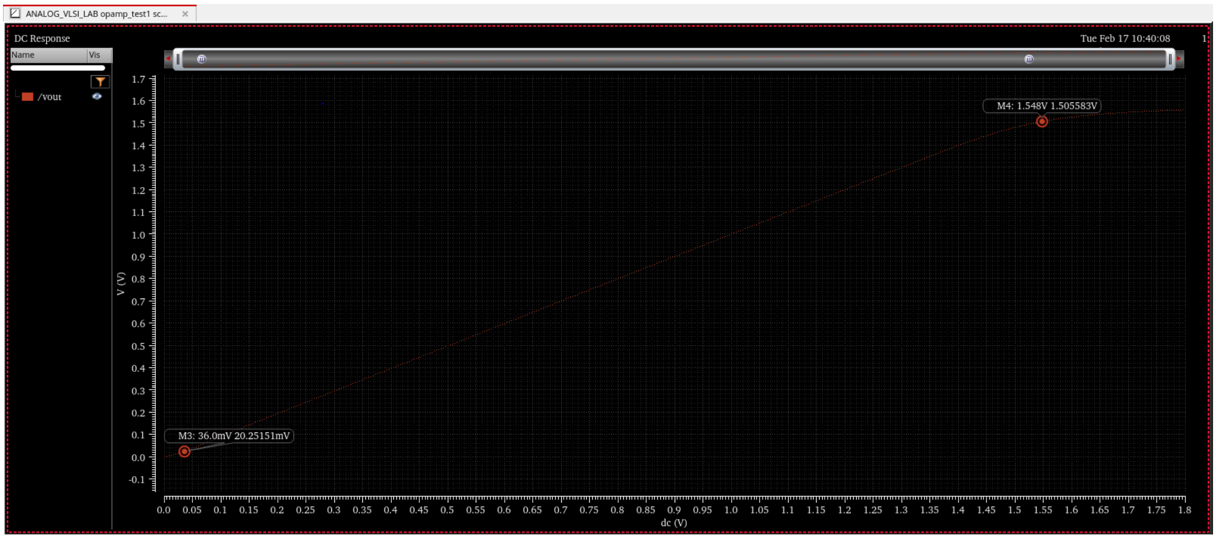

Figure 18: ICMR for our given circuit

The input common-mode range (ICMR) defines the allowable range of DC input voltage over which the operational amplifier maintains proper biasing and stable operation while preserving the required gain. shows the variation of differential gain with respect to the applied common-mode input voltage.

From the plot, the amplifier maintains a gain of approximately 60 dB within the input voltage range of 20.2515m V to 1.5055 V, which corresponds to the valid ICMR. Below the lower limit, the input differential pair enters the triode region or cuts off, whereas above the upper limit, the current mirror and load devices move out of saturation, resulting in gain degradation.

This analysis verifies that the designed amplifier satisfies the required input swing constraints while ensuring stable biasing and high gain operation. Proper ICMR is essential for maximizing signal swing, preventing distortion, and maintaining linearity in precision analog applications.

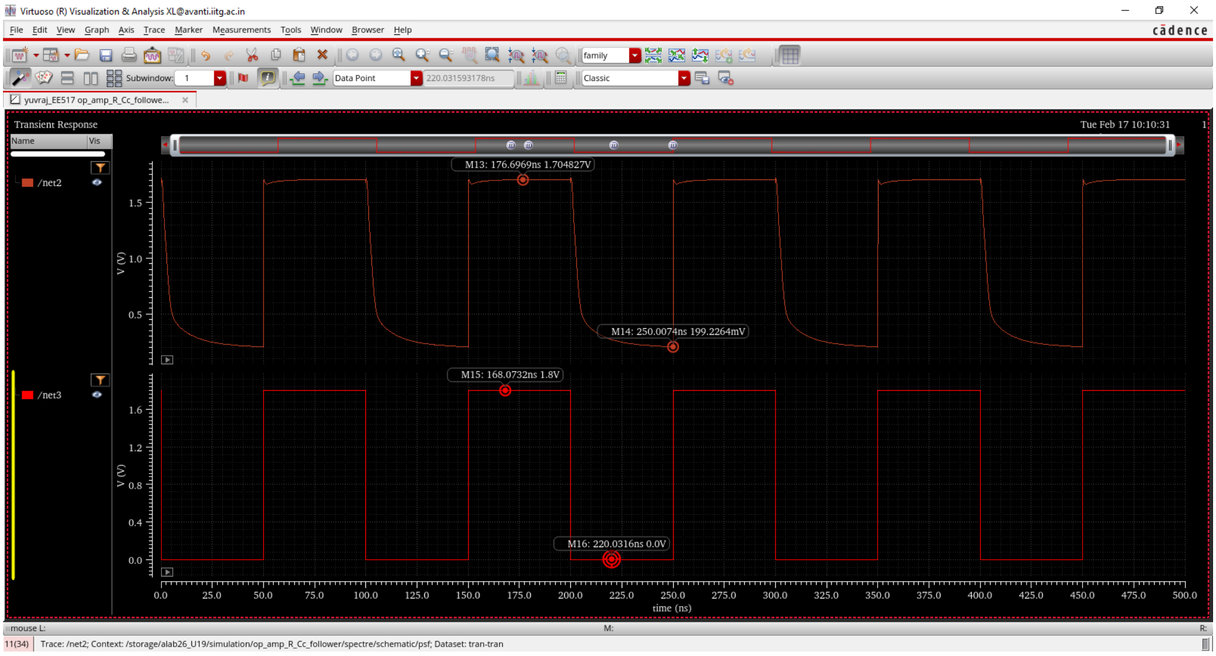

Figure 19: OCMR for 2 stage op amp with coupling capacitance and resistance

The output common-mode range (OCMR) defines the maximum allowable voltage swing at the output node for which the operational amplifier maintains linear operation without distortion or saturation. shows the transient response of the amplifier configured as a voltage follower when excited with a pulse input varying from 0 to 1.8 V.

From the observed output waveform, the amplifier is able to swing between approximately 0.199 V and 1.705 V, which represents the valid OCMR. The lower and upper swing limitations arise due to saturation of the output-stage transistors and headroom constraints imposed by the bias currents and device overdrive voltages.

This analysis confirms that the amplifier supports a wide output voltage swing within the supply limits, ensuring low distortion and reliable large-signal operation. Proper OCMR is critical for maximizing dynamic range and maintaining linearity in high-performance analog circuits.

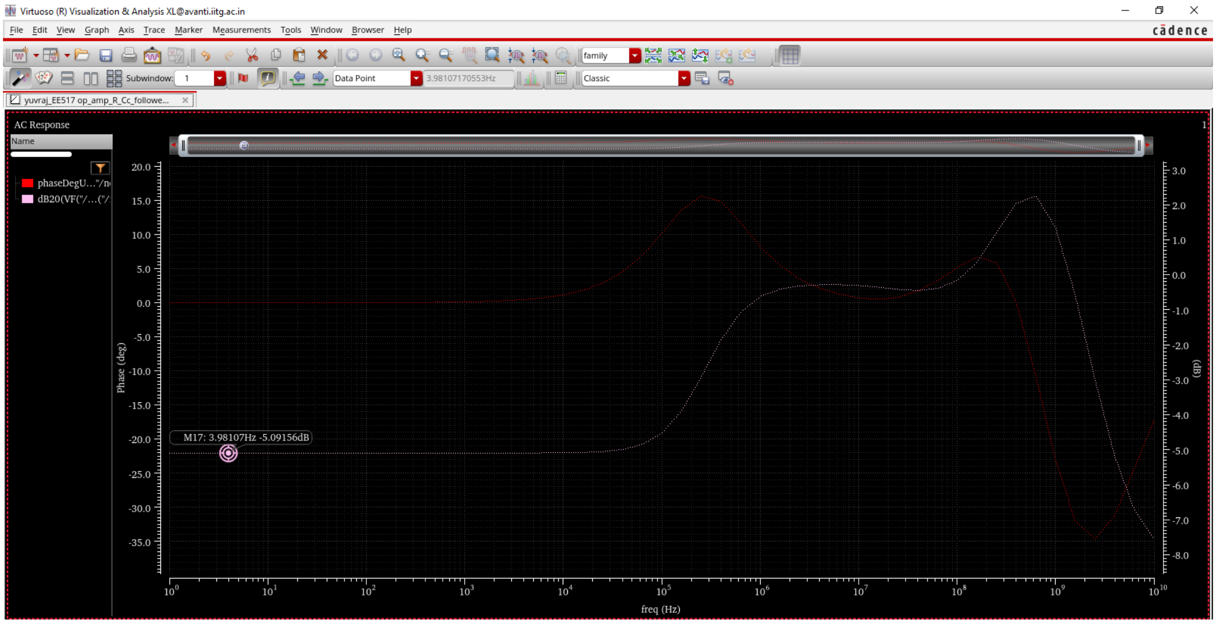

Figure 20: PSRR for 2 stage op amp with coupling capacitance

The power supply rejection ratio (PSRR) characterizes the ability of an operational amplifier to suppress variations in the supply voltage from appearing at the output. It is defined as

$$\mathrm{PSRR}(f) = 20 \log_{10} \left(\frac{A_d(f)}{A_{ps}(f)} \right), \tag{23}$$

where $A_d(f)$ is the differential gain and $A_{ps}(f)$ is the gain from supply to output. In this analysis, a 10 mV AC signal is applied at V_{DD}, and the resulting output response is used to compute the common-mode gain, from which PSRR is derived.

At low frequencies, the PSRR is high due to the strong isolation provided by the biasing current sources and high output resistance of the active devices, effectively attenuating supply noise. As frequency increases, parasitic capacitances and finite output impedance introduce direct coupling paths between the supply and output nodes, resulting in degradation of PSRR. The mid-frequency peak observed in the response is attributed to pole–zero interactions introduced by the compensation capacitor and internal node capacitances.

At high frequencies, the PSRR further deteriorates due to reduced loop gain and dominant capacitive feedthrough, allowing more supply noise to propagate to the output. This behavior highlights the inherent trade-off between bandwidth and supply noise rejection in high-speed operational amplifier design and emphasizes the importance of compensation and layout optimization for robust power supply immunity.

5 Conclusion

Table 1 summarizes the key performance parameters of the two-stage operational amplifier implemented with only Miller compensation capacitor (C_c) and with resistor–capacitor compensation (R–C_c). The comparison highlights the impact of compensation strategy on stability, bandwidth, dynamic response, and operating ranges.

The observed performance differences between the two compensation techniques arise

Table 1: Performance Comparison of Two-Stage Op-Amp with C_c and R–C_c Compensation

S. No.	**Parameter**	**Only C_c**	R–C_c
1	DC Gain (dB)	60	72.47
2	Unity Gain Frequency (UGF)	210 Mhz	200Mhz
3	Phase at UGF (deg)	56.46 deg	61.716 deg
4	Slew Rate rising (Low Frequency)	$180 * 10^6$	$2560 * 10^6$
5	Slew Rate rising (UBW)	$109 * 10^6$	$1391 * 10^6$
6	Slew Rate rising (High Frequency)	N/A	$388 * 10^6$
7	Slew Rate falling (Low Frequency)	$-110 * 10^6$	$-300 * 10^6$
8	Slew Rate falling (UBW)	$-106 * 10^6$	$-323 * 10^6$
9	Slew Rate falling (High Frequency)	N/A	$-401.6 * 10^6$
10	Input Common-Mode Range (ICMR)	0.68V to 1.35V	20.2515mV to 1.5055mV
11	Output Common-Mode Range (OCMR)	0.27V to 1.57V	0.199V to 1.7V
12	PSRR	-11.8dB	-5.6dB
13	Common Mode Gain	-11.338dB	-19.3701dB

primarily from their impact on frequency response and stability. In the C_c-only compensated amplifier, the Miller capacitor introduces a dominant pole at low frequency, improving stability but significantly reducing bandwidth and slew rate due to increased effective capacitance at the first-stage output. This results in lower unity-gain frequency and slower transient response.

In contrast, the introduction of a series resistor with C_c generates a left-half-plane zero, which compensates for the phase lag caused by the non-dominant pole. This improves phase margin, increases bandwidth, and enhances slew rate by reducing the effective Miller multiplication. Consequently, the R–C_c compensated amplifier exhibits superior frequency response, faster settling behavior, and improved dynamic performance while maintaining adequate stability.

Overall, the R–C_c compensation technique provides a better trade-off between stability, bandwidth, and transient response, making it more suitable for high-speed and wideband analog applications.